

Normative data on the daily profile of urinary 6-sulfatoxymelatonin in healthy subjects between the ages of 20 and 84.

Although the hormone melatonin is a key factor for the proper functioning of the circadian timing system (CTS) and exogenous melatonin has been shown to be beneficial in cases of CTS disturbances, a deficit of melatonin has yet to be defined as a disorder. The aim of our study was to collect a normative data set on 24-h melatonin excretion in healthy human adults living in a natural environment. Urine samples were collected from 75 healthy subjects (45 women/30 men; mean age 47.2, SD 19.5, range 20-84) after five consecutive periods: 2300-0700, 0700-1100, 1100-1800, 1800-2300 and 2300-0700 h. 6-Sulfatoxymelatonin (aMT6s) concentrations were analyzed in duplicate by IBL (Hamburg) using a highly sensitive, competitive ELISA kit. Twenty-four hour-aMT6s total amount ($\rho=-0.68$, $p<0.001$), aMT6s nighttime excretion ($\rho=-0.69$, $p<0.001$), aMT6s morning excretion ($\rho=-0.66$, $p<0.001$) and evening excretion ($r=-0.26$, $p=0.023$) were negatively associated with age, whereas daytime excretion ($r=-0.17$, $p=0.15$) was not. The intra-subject night-day ratio varied up to 10.5 (mean 6.0) in young subjects (aged 20-35) and up to 5.4 (mean 2.8) in older individuals (age>65). The total amount of 24 h-aMT6s (range 7.5-58 microg) as well as the amount of aMT6s excreted during the nighttime period (range 327-6.074 ng/h) varied as much as 20-fold between individuals. Our data show an age-related decline in melatonin excretion in healthy subjects living in a natural environment. The high inter-individual variability of excretion rates may explain why a normative data set is of no use in replacement strategies.